

UNIT-1 RELATIONS AND FUNCTIONS

CASE BASED QUESTIONS

1.

- (a) Students of a school are taken to a railway museum to learn about railways heritage and its history.



An exhibit in the museum depicted many rail lines on the track near the railway station. Let L be the set of all rail lines on the railway track and R be the relation on L defined by

$$R = \{(l_1, l_2) : l_1 \text{ is parallel to } l_2\}$$

On the basis of the above information, answer the following questions :

- (i) Find whether the relation R is symmetric or not.
- (ii) Find whether the relation R is transitive or not.
- (iii) If one of the rail lines on the railway track is represented by the equation $y = 3x + 2$, then find the set of rail lines in R related to it.

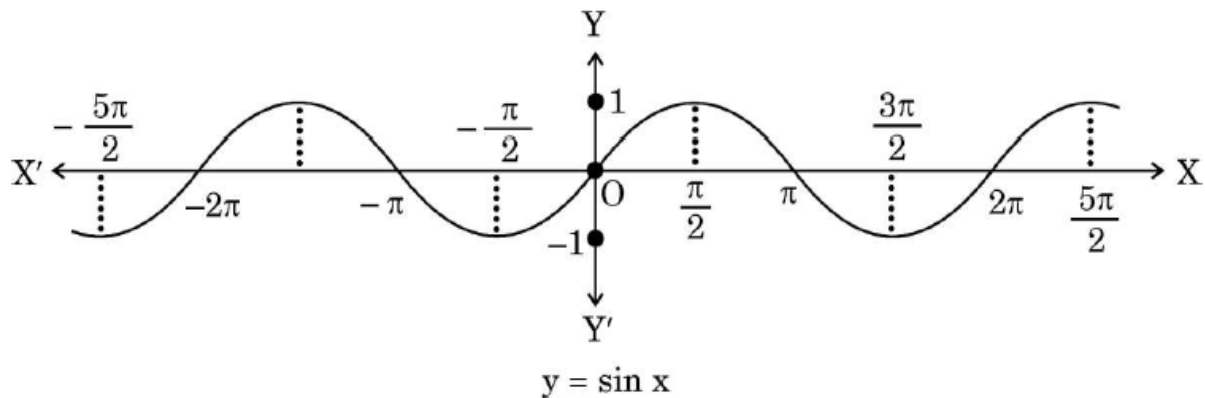
OR

- (b) Let S be the relation defined by $S = \{(l_1, l_2) : l_1 \text{ is perpendicular to } l_2\}$ check whether the relation S is symmetric and transitive.

2.

If a function $f : X \rightarrow Y$ defined as $f(x) = y$ is one-one and onto, then we can define a unique function $g : Y \rightarrow X$ such that $g(y) = x$, where $x \in X$ and $y = f(x)$, $y \in Y$. Function g is called the inverse of function f .

The domain of sine function is $\mathbb{R}$ and function $\sin : \mathbb{R} \rightarrow \mathbb{R}$ is neither one-one nor onto. The following graph shows the sine function.



Let sine function be defined from set A to $[-1, 1]$ such that inverse of sine function exists, i.e., $\sin^{-1} x$ is defined from $[-1, 1]$ to A .

On the basis of the above information, answer the following questions :

- (i) If A is the interval other than principal value branch, give an example of one such interval.
- (ii) If $\sin^{-1}(x)$ is defined from $[-1, 1]$ to its principal value branch, find the value of $\sin^{-1}\left(-\frac{1}{2}\right) - \sin^{-1}(1)$.
- (iii) (a) Draw the graph of $\sin^{-1} x$ from $[-1, 1]$ to its principal value branch.

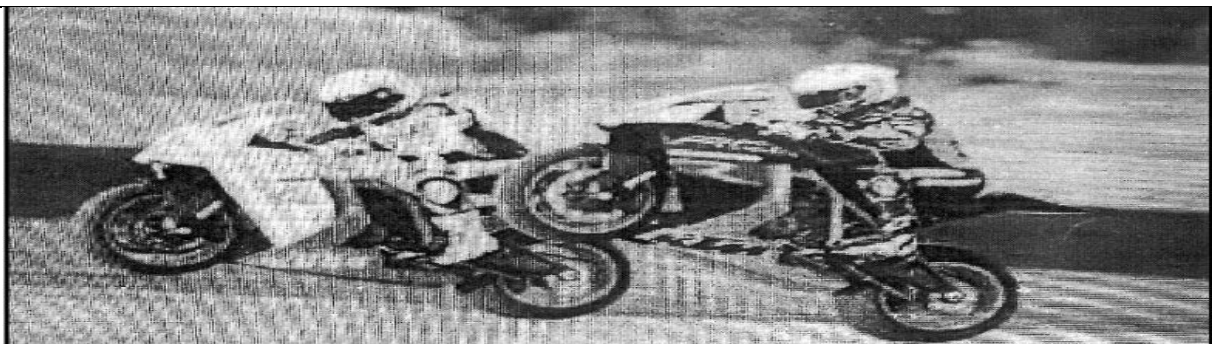
OR

- (iii) (b) Find the domain and range of $f(x) = 2 \sin^{-1}(1 - x)$.

3.

An organization conducted bike race under two different categories – Boys and Girls. There were 28 participants in all. Among all of them, finally three from category 1 and two from category 2 were selected for the final race. Ravi forms two sets B and G with these participants for his college project.

Let $B = \{b_1, b_2, b_3\}$ and $G = \{g_1, g_2\}$, where B represents the set of Boys selected and G the set of Girls selected for the final race.



Based on the above information, answer the following questions :

- (I) How many relations are possible from B to G ?
- (II) Among all the possible relations from B to G, how many functions can be formed from B to G ?
- (III) Let $R : B \rightarrow B$ be defined by $R = \{(x, y) : x \text{ and } y \text{ are students of the same sex}\}$. Check if R is an equivalence relation.

OR

- (III) A function $f : B \rightarrow G$ be defined by $f = \{(b_1, g_1), (b_2, g_2), (b_3, g_1)\}$.

Check if f is bijective. Justify your answer.

- 4.** A class-room teacher is keen to assess the learning of her students the concept of “relations” taught to them. She writes the following five relations each defined on the set $A = \{1, 2, 3\}$:

$$R_1 = \{(2, 3), (3, 2)\}$$

$$R_2 = \{(1, 2), (1, 3), (3, 2)\}$$

$$R_3 = \{(1, 2), (2, 1), (1, 1)\}$$

$$R_4 = \{(1, 1), (1, 2), (3, 3), (2, 2)\}$$

$$R_5 = \{(1, 1), (1, 2), (3, 3), (2, 2), (2, 1), (2, 3), (3, 2)\}$$

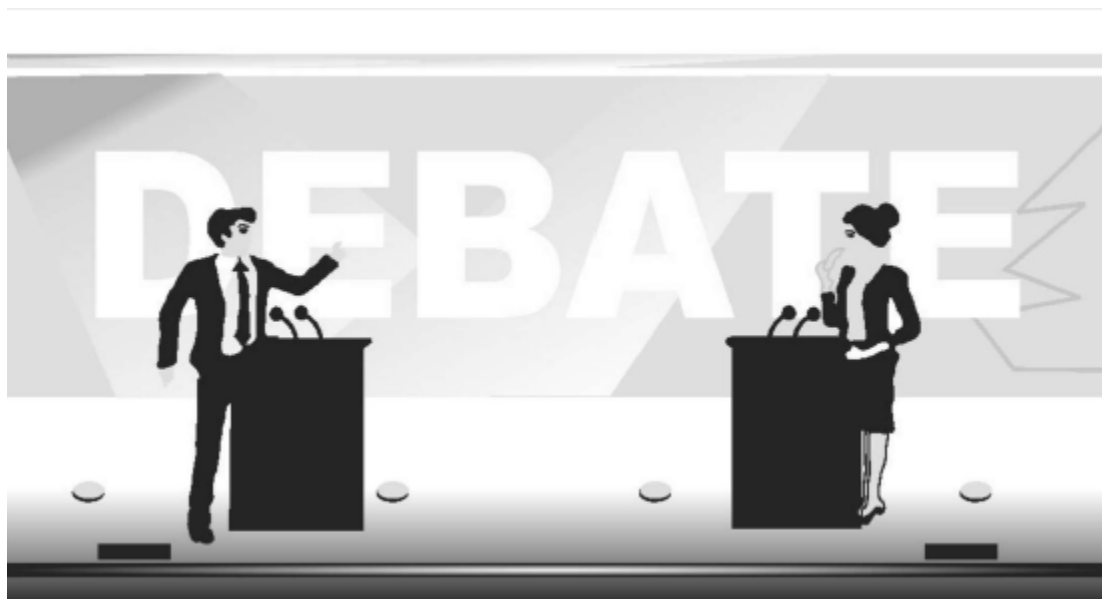
The students are asked to answer the following questions about the above relations :

- (i) Identify the relation which is reflexive, transitive but not symmetric.
- (ii) Identify the relation which is reflexive and symmetric but not transitive.
- (iii) (a) Identify the relations which are symmetric but neither reflexive nor transitive.

OR

- (iii) (b) What pairs should be added to the relation R_2 to make it an equivalence relation ?

5. A school is organizing a debate competition with participants as speakers $S = \{S_1, S_2, S_3, S_4\}$ and these are judged by judges $J = \{J_1, J_2, J_3\}$. Each speaker can be assigned one judge. Let R be a relation from set S to J defined as $R = \{(x, y) : \text{speaker } x \text{ is judged by judge } y, x \in S, y \in J\}$.



Based on the above, answer the following :

- (i) How many relations can be there from S to J ? 1
- (ii) A student identifies a function from S to J as $f = \{(S_1, J_1), (S_2, J_2), (S_3, J_2), (S_4, J_3)\}$ Check if it is bijective. 1
- (iii) (a) How many one-one functions can be there from set S to set J ? 2

OR

- (iii) (b) Another student considers a relation $R_1 = \{(S_1, S_2), (S_2, S_4)\}$ in set S . Write minimum ordered pairs to be included in R_1 so that R_1 is reflexive but not symmetric. 2

6. Let A be the set of 30 students of class XII in a school. Let $f : A \rightarrow N$, N is a set of natural numbers such that function $f(x) = \text{Roll Number of student } x$. On the basis of the given information, answer the following :
- (i) Is f a bijective function ? 1
- (ii) Give reasons to support your answer to (i). 1

	(iii) (a) Let R be a relation defined by the teacher to plan the seating arrangement of students in pairs, where $R = \{(x, y) : x, y \text{ are Roll Numbers of students such that } y = 3x\}$. List the elements of R. Is the relation R reflexive, symmetric and transitive ? Justify your answer. 2 OR (iii) (b) Let R be a relation defined by $R = \{(x, y) : x, y \text{ are Roll Numbers of students such that } y = x^3\}$. List the elements of R. Is R a function ? Justify your answer. 2
7.	An organization conducted bike race under 2 different categories-boys and girls. In all, there were 250 participants. Among all of them finally three from Category 1 and two from Category 2 were selected for the final race. Ravi forms two sets B and G with these participants for his college project. Let $B = \{b_1, b_2, b_3\}, G = \{g_1, g_2\}$ where B represents the set of boys selected and G the set of girls who were selected for the final race. Ravi decides to explore these sets for various types of relations and functions. On the basis of the above information, answer the following questions: (i) Ravi wishes to form all the relations possible from B to G. How many such relations are possible? [1Mark] (ii) Write the smallest equivalence relation on G. [1Mark] (iii) (a) Ravi defines a relation from B to B as $R_1 = \{(b_1, b_2), (b_2, b_1)\}$. Write the minimum ordered pairs to be added in R_1 so that it becomes (A) reflexive but not symmetric, (B) reflexive and symmetric but not transitive. [2 Marks] OR (iii) (b) If the track of the final race (for the biker b_1) follows the curve $x^2 = 4y$; (where $0 \leq x \leq 20\sqrt{2}$ & $0 \leq y \leq 200$), then state whether the track represents a one-one and onto function or not. (Justify). [2 Marks]

8. A city's traffic management department is planning to optimize traffic flow by analyzing the connectivity between various traffic signals. The city has five major spots labelled A, B, C, D , and E .



The department has collected the following data regarding one-way traffic flow between spots:

1. Traffic flows from A to B , A to C , and A to D .
2. Traffic flows from B to C and B to E .
3. Traffic flows from C to E .
4. Traffic flows from D to E and D to C .

The department wants to represent and analyze this data using relations and functions. Use the given data to answer the following questions:

- I. Is the traffic flow reflexive? Justify. [1]
- II. Is the traffic flow transitive? Justify. [1]
- III A. Represent the relation describing the traffic flow as a set of ordered pairs. Also state the domain and range of the relation.

OR

- III B. Does the traffic flow represent a function? Justify your answer. [2]

